

HR People Analytics Dashboard

Technical Documentation

Prepared by: Omar Magdy
Tool: Microsoft Power BI Desktop
File: HR_Dashboard.pbix
Last Updated: May 2026

1. Project Overview

The HR People Analytics Dashboard is a fully interactive Microsoft Power BI report designed to give HR managers and business leaders a real-time, data-driven view of their workforce. It consolidates headcount, salary, turnover, and performance data into a single, filterable interface — eliminating manual reporting and enabling faster, evidence-based decisions.

1.1 Key Capabilities

- Real-time headcount tracking with month-over-month and year-over-year comparisons
- Salary benchmarking by department, gender, and salary band
- Turnover rate analysis at the department level via a waterfall chart
- Interactive filters for department, gender, salary range, performance rating, and date range
- Custom HTML visuals for animated KPI cards, sidebar navigation, and tooltip enhancements
- Custom Power BI theme applied throughout for consistent branding

2. Report Structure

The report contains three pages, each serving a distinct purpose:

Page Name	Type	Purpose
Main Dashboard	Primary Report Page	Central workspace housing all KPI cards, charts, slicers, and navigation elements visible to end users.
Date Chart Tool Tip	Tooltip Page	Hidden page used as a custom tooltip for the Active Employees area chart, showing date-level headcount details on hover.

Page 1	Overlay Page	Secondary page hosting the collapsible filter panel overlay triggered from the main dashboard.
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3. Data Model

3.1 Tables

The data model is built on two core tables:

Table Name	Type	Description
Dim_Employee_Master	Dimension / Fact	Main employee table. Contains one row per employee with all demographic, role, salary, and employment lifecycle columns.
Dim_Calendar	Date Dimension	Standard calendar table linked to Dim_Employee_Master via HireDate and TerminationDate. Supports time-intelligence DAX.

3.2 Key Columns — Dim_Employee_Master

Column	Data Type	Description
HireDate	Date	Date the employee was hired. Used to compute headcount at any point in time.
TerminationDate	Date	Date of termination. Blank for active employees.
Status	Text	Employment status. Values include 'Active' and 'Terminated'.
Department	Text	Employee's department. Used across all department-level breakdowns.
Gender	Text	Employee gender. Used in salary and headcount gender analysis.
Salary Band	Text	Salary bracket grouping (e.g. Low, Mid, High). Drives salary distribution chart.
PerformanceRating	Text	Performance score or category. Used by the Performance Rating slicer.

3.3 Key Columns — Dim_Calendar

Column	Data Type	Description
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Date	Date	Primary date key. Linked to HireDate and TerminationDate in Dim_Employee_Master.
Year	Integer	Calendar year extracted from Date.
Quarter	Text	Calendar quarter (Q1–Q4).
Month	Text	Month name or number. Used in time-series charts and tooltip drilldowns.
Day	Integer	Day of month. Used in the date hierarchy for detailed drill-down.

4. DAX Measures

All measures are defined on the Dim_Employee_Master table. The following table documents each measure, its business purpose, and the underlying logic.

Measure Name	Category	Business Logic
Headcount at Date	Headcount	Returns the number of active employees at a selected date. An employee is counted if their HireDate is on or before the selected date AND their TerminationDate is either blank or after the selected date.
Active Employees	Headcount	Aggregate measure displaying total currently active employees. Powers the KPI cards and area chart.
MoM Change %	Trend	Month-over-month percentage change in headcount. Compares Headcount at Date for the selected month against the previous month using EDATE. Returns BLANK if no prior month data exists.
YoY Change %	Trend	Year-over-year percentage change in headcount. Compares current Headcount at Date against the same date 12 months prior using EDATE.
New Hires in Month	Movement	Count of employees whose HireDate falls within the selected calendar month (start of month to end of month via EOMONTH).
Terminations in Month	Movement	Count of employees with a TerminationDate within the selected month whose Status is 'Terminated'.
Net Change in Month	Movement	Simple difference: New Hires in Month minus Terminations in Month. Shows net workforce growth or decline for the period.

Last 6 Months HC	Trend	Rolling headcount over the last 6 months using DATESINPERIOD. Designed for sparkline use in tooltip visuals.
Average Salary	Compensation	Average salary across filtered employees. Used in department and gender salary comparison charts.
Count Of Employee	Headcount	Raw count of employee rows in the current filter context. Used in the salary distribution column chart.
Termination Rate by Dept	Attrition	Turnover rate per department. Calculated as terminated employees divided by total employees in each department. Drives the waterfall chart.
Top Dept Name	Insight	Returns the name of the department with the highest headcount at the selected date using TOPN + SUMMARIZE.
Top Dept Count	Insight	Returns the headcount of the top department identified by Top Dept Name.

5. Visuals Inventory

The Main Dashboard page contains the following visual components:

Visual Title	Visual Type	Fields Used	Purpose
HR KPI Cards	Custom HTML Visual	HR KPI Cards (measure)	Animated KPI card panel showing headline workforce metrics such as active headcount, MoM change, and YoY change.
Active Employees (Trend)	Area Chart	Date Hierarchy, Active Employees	Time-series line showing workforce headcount over Year / Quarter / Month / Day with drill-down capability.
Active Employees by Department Top 5	Bar Chart	Department, Active Employees	Horizontal bar chart ranking the top 5 departments by current active headcount.
Average Salary by Department Top 5	Column Chart	Department, Average Salary	Column chart comparing average salary across the top 5 departments.
Salary Distribution	Column Chart	Salary Band, Count Of Employee	Distribution of employees across salary bands, showing workforce concentration.

Average Salary by Gender	Column Chart	Gender, Average Salary	Side-by-side salary comparison by gender for equity analysis.
Turnover Rate by Department	Waterfall Chart	Department, Termination Rate	Waterfall chart breaking down turnover rate contribution by department.
Nav Bar	Custom HTML Visual	HR Right Sidebar Nav	Collapsible right-side navigation panel for switching between report pages or views.
Animated Logo	Custom HTML Visual	HR Animated Logo	Branded animated header logo for visual polish.
Filter Panel	Custom HTML Visual	Filter Panel	Overlay filter panel activated by a button, providing a clean UX for applying multiple slicers at once.
Date Chart Tool Tip	Custom HTML Visual	Line Chart Tooltip	Custom tooltip page shown on hover over the area chart, displaying Headcount at Date, MoM %, YoY %, new hires, and terminations.

6. Slicers & Interactive Filters

The dashboard provides five slicers that allow users to cross-filter all visuals simultaneously:

Slicer	Source Field	Effect
Department	<code>Dim_Employee_Master[Department]</code>	Filters all visuals to show data for one or more selected departments.
Gender	<code>Dim_Employee_Master[Gender]</code>	Filters workforce data by gender, enabling equity and pay gap analysis.
Salary Range	<code>Dim_Employee_Master[Salary Band]</code>	Limits the view to employees within selected salary bands.
Performance Rating	<code>Dim_Employee_Master[PerformanceRating]</code>	Filters by employee performance score, useful for high-performer retention analysis.
Date Range	<code>Dim_Calendar[Date]</code>	Controls the time window for all time-based calculations including headcount trends and MoM/YoY metrics.

7. Custom Visuals & Theme

7.1 Custom Visuals Used

Two third-party / custom HTML visuals are used in this report:

Visual ID	Usage
htmlContent443BE3AD5...	Multi-purpose HTML visual used for: KPI Cards, Animated Logo, Sidebar Navigation, Filter Panel, and the Line Chart Tooltip. Renders custom HTML/CSS/JS inside the Power BI canvas.
histogramF963F133BA...	Histogram custom visual — available in the report package for optional salary or age distribution visualisation.

7.2 Theme

- Theme file: Power_BI_Theme_Generator_-_BIB3670118859646093.json
- Applied as a registered resource, ensuring consistent colours, fonts, and visual formatting across all report pages.
- Base theme: CY26SU04 (Power BI default base, overridden by the custom theme).

8. How to Use the Dashboard

8.1 Requirements

- Microsoft Power BI Desktop (latest version recommended)
- Access to the source data (employee master table and calendar table)
- Custom visuals (.pbviz files) are bundled inside the .pbix — no separate installation needed

8.2 Connecting Your Data

The report is built on a Dim_Employee_Master table and a Dim_Calendar table. To use with your own data:

1. Open HR_Dashboard.pbix in Power BI Desktop.
2. Go to Home > Transform Data to open Power Query Editor.
3. Replace the sample data source with your own CSV, Excel, SQL, or API connection.
4. Ensure column names match those listed in Section 3 (or remap them in Power Query).
5. Click Close & Apply, then Refresh to load your data.

8.3 Using the Filters

- Use the slicers on the right-hand panel to filter by Department, Gender, Salary Range, Performance Rating, or Date Range.
- Click the filter icon button to open or close the collapsible Filter Panel overlay for a cleaner filtering experience.
- All visuals respond to slicer selections simultaneously — no manual refresh needed.
- Hover over the Active Employees area chart to see the custom tooltip showing Headcount at Date, MoM %, YoY %, New Hires, and Terminations.

9. Technical Notes

9.1 Time Intelligence

All time-based measures use Dim_Calendar[Date] as the anchor. Ensure the calendar table covers the full range of HireDate and TerminationDate values in your data. Gaps in the calendar table will produce BLANK values in MoM % and YoY % measures.

9.2 Headcount Logic

The Headcount at Date measure uses the following inclusion rule:

- Employee HireDate <= Selected Date
- Employee TerminationDate > Selected Date OR TerminationDate is BLANK

This ensures both active employees (blank termination date) and those terminated after the selected date are correctly excluded.

9.3 Top N Filtering

The 'Top 5' charts (Active Employees by Department, Average Salary by Department) use Power BI's built-in Top N visual filter — not a DAX measure — applied at the visual level. This means the ranking dynamically updates when slicers are applied.

9.4 Custom HTML Visuals

The HTML content visuals render client-side HTML, CSS, and JavaScript. They require the 'Allow visuals created using the Power BI SDK' setting to be enabled in Power BI Desktop options. In Power BI Service (cloud), ensure the organisational setting for custom visuals is enabled by the tenant admin.

10. Contact & Support

This dashboard was designed and developed by Omar Magdy, Power BI Developer & Data Analyst.

Channel	Details
Upwork	upwork.com/freelancers/~0166cc6ba4e7d3f5c7
Portfolio	omarmagdy99.github.io/omar-magdy-portfolio
HR Dashboard	omarmagdy99.github.io/hr-analytics-dashboard

For questions about data connections, DAX customisation, or extending the dashboard with additional pages or visuals, feel free to reach out via Upwork.